AN EXACT t-TEST

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Multivariate linear regression and randomization-based inference are two essential methods in statistics and econometrics. Nevertheless, the problem of producing a randomized test for the value of a single regression coefficient that is exactly valid when errors are exchangeable, and which is asymptotically valid for the best linear predictor, has remained elusive. In this paper, we produce a test that is exactly valid with exchangeable errors and which allows for general covariate designs; covariates may be continuous as well as discrete and may be correlated. The test is asymptotically valid when the errors are not exchangeable, in particular in the presence of conditional heteroskedasticity.

KEYWORDS: Randomization inference, linear regression.

1. INTRODUCTION

Consider the linear regression model

$$Y_i = \beta_0 + X_{i1}\beta_1 + \cdots + X_{ip}\beta_p + \varepsilon_i,\ i = 1, \ldots, n, \tag{1}$$

where $\{Y_i, X_i, \varepsilon_i\}_{i=1}^n$is drawn from an unknown distribution $\mathcal{P}_n$. For ease of exposition, we will consider the vector representation of the bivariate model,

$$\mathbf{Y} = \mathbf{1}_n\beta_0 + \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \boldsymbol{\varepsilon}, \tag{2}$$

where $\mathbf{1}_n$ is a $n \times 1$ vector of 1's, $\mathbf{X}_j = (X_{1j}, \ldots, X_{nj})^T$, $j = 1, 2$, and $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)$. All results and manipulations can be done with minor modifications for matrices of general dimensions. In particular, the reader may think of $\mathbf{X}_2$ as being an $n \times (p-1)$ matrix without any formal incongruities, and all technical results are proved for that case.

Our interest lies in testing the value of the first slope coefficient, i.e., in testing the null

$$H_0 : \beta_1 = \beta_1^0. \tag{3}$$

Importantly, inverting such a test produces a marginally valid confidence interval for β_1. Of course, the first covariate is chosen without loss of generality since the order of the covariates does not impact the value of their estimate.

A test is *valid* if its probability of false rejection does not exceed the user-prescribed, nominal probability $\alpha \in [0, 1]$. It is *exact* if it is valid for any given sample size, and it has *nominal size* if it has probability of rejection exactly α under the null.

The standard approach for testing (3) is to use the classical t-test (Student, 1908, Wooldridge, 2010). This test is asymptotically valid and of nominal size under standard conditions. In the

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event that the errors are independent and identically distributed Gaussian random variables, the t-test will be exact and of nominal size.

Another classical approach to testing is randomization. Indeed, in applications such as one- or two-sample tests for the mean, tests of economic inequality (Dufour et al., 2019), as well as many other examples (Romano, 1990, Lehmann and Romano, 2005), randomization-based approaches deliver exact tests without making assumptions about the specific distribution of the data. To the extent that such tests have large-sample performance comparable to that of standard tests under comparable assumptions, they may provide a desirable alternative requiring weaker assumptions for exact validity.

The question of producing an exactly valid, randomization-based test for (3) that is asymptotically valid under general conditions thus arises very naturally.

However, Lei and Bickel (2021) provide an extensive literature review substantiating the claim that, for null hypotheses such as (3) , *"there is no test that is exact under reasonably general assumptions*". Their notion of "general assumptions" rules out covariate designs where all nuisance covariates are discrete. They presumably rule out $\max$-p procedures as too conservative,[1] or are restricting themselves to exact tests of nominal size.

Thus motivated, Lei and Bickel (2021) produced the first such exact test for general linear hypotheses in the linear model (1). They use a construction that cleverly builds the required invariances into the test statistic. See Section 3 of the Supplementary Appendix.

Wen et al. (2022) later produced another such test which is better adapted to high-dimensional regression and which, although it does not have nominal size —i.e., it rejects under the null hypothesis with probability lower than the nominal level α— in some cases displays better power properties. They use the idea of somehow explicitly "orthogonalizing" permuted nuisance regressors. This idea, while it is instantiated quite differently here, plays a prominent role in the proposed method and is inspired by their approach.

While these procedures closed an important mathematical statistics question, neither of these are shown to be asymptotically valid when exchangeability does not avail, for instance under heteroskedasticity —indeed, they are not, see the Supplementary Appendix. This is critical for applications since in practice we typically cannot assume that the conditional expectation is linear, but are content with estimating its best approximation, the best linear predictor $E^*[Y_i|X_i] = X_i^T\beta$, $X_i = (1, X_{i1}, \ldots, X_{ip})$, whose errors $\varepsilon_i = Y_i - E^*[Y_i|X_i]$ satisfy orthogonality conditions $E[\varepsilon_i] = 0$ and $E[\varepsilon_i \cdot X_{ij}] = 0$, $j = 1, \ldots, p$. It is furthermore often implausible that the thus produced errors retain no dependence with the covariates. In addition, it effectively rules out the more general best linear predictor interpretation we sought in the first place. Heterogeneous treatment effects, for instance, bring about heteroskedasticity (Breusch and Pagan, 1979).

As it pertains to producing such an asymptotically robust randomization test, the case in which the tested covariate is independent of other covariates is covered in DiCiccio and Romano (2017) and substantially generalized in Young (2023).

The first and only exactly valid procedure allowing for dependent covariates that is also asymptotically valid for the best linear predictor under heteroskedasticity was produced by D'Haultfœuille and Tuvaandorj (2024). Their work builds on the studentization strategy of DiCiccio and Romano (2017) and others cited therein. Their method applies to the case in which all nuisance covariates are discrete and enough support points of the nuisance covariates

[1] Given a procedure producing a p-value $p(\beta_1, \beta_2, \ldots, \beta_p)$ for null hypotheses on the full vector $(\beta_1, \beta_2, \ldots, \beta_p)$, we speak of $\max_{\beta_2,\ldots,\beta_p} p(\beta_1, \beta_2, \ldots, \beta_p)$ as the max-p statistic, a conservative yet valid p-value corresponding to the projection method. Because they take a worst case over nuisance parameters, we do not consider $\max$-p statistics to be "inherently" invariant with respect to nuisance parameters.

vector are sampled an asymptotically growing number of times. They state that "*constructing a permutation test for subvectors that is both exact under independence and asymptotically heteroskedasticity-robust for any design*" is yet unresolved.

Our contribution is to produce such a test. Let $\mathbf{G}_n$ be any group of permutations $g : \{1,\dots,n\} \to \{1,\dots,n\}$. For a given vector $\mathbf{Z} \in \mathbb{R}^n$, equivalently write $g\mathbf{Z}$ or $\mathbf{Z}_g$ to designate $(Z_{g(1)},\dots,Z_{g(n)})^T$, the vector whose entries are permuted according to g.[2]

We propose the test statistic

$$t_g\left(\beta_1^0\right) = \frac{\overline{\mathbf{X}}_1^T\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)_g}{\hat{\sigma}_g}, \tag{4}$$

for each $g \in \mathbf{G}_n$, with

$$\hat{\sigma}_g^2 = \frac{1}{n}\sum_{i=1}^{n} \overline{\mathbf{X}}_{i1}^2 \hat{\varepsilon}_{g(i)}^2, \tag{5}$$

where $\overline{\mathbf{X}}_1$ is the vector of fitted residuals from ordinary least-squares regression of $\mathbf{X}_1$ on an intercept term and all permitted permutations of the nuisance regressor, i.e., $\{g\mathbf{X}_2\}_{g\in\mathbf{G}_n} \equiv \{(X_{g(1),2},\dots,X_{g(n),2})\}_{g\in\mathbf{G}_n}$. The fitted residuals $\hat{\varepsilon}$ obtain from ordinary least-squares regression of $\mathbf{Y}$ on an intercept term, $\{g\mathbf{X}_1\}_{g\in\mathbf{G}_n}$ and $\{g\mathbf{X}_2\}_{g\in\mathbf{G}_n}$. By orthogonalizing $g\mathbf{X}_2$ for all $g \in \mathbf{G}_n$, each t_g is invariant with respect to β_2. See Section 2.1 below. As detailed in Section 2, we suggest using a group of non-overlapping block permutations as the group of actions $\mathbf{G}_n$.

Intuitively, $\overline{\mathbf{X}}_1^T\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)$ measures the variation due to X_1 left in the "short residuals" $\mathbf{Y} - \mathbf{X}_1\beta_1^0$ and captured by $\overline{\mathbf{X}}_1$, a version of $\mathbf{X}_1$ from which we extracted variation due to $\mathbf{X}_2$ and its block permutations. Using $\overline{\mathbf{X}}_1$ in lieu of $\mathbf{X}_1$ diminishes power but makes the procedure invariant, which is a requirement for validity. When $g \neq \mathrm{id}$, the null is "enforced" since $\overline{\mathbf{X}}_1$ will typically pick up very little X_1 variation from the permuted short residuals, $\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)_g$. For, say, a one-sided test, one computes the p-value as the quantile of $t_{\mathrm{id}}\left(\beta_1^0\right)$ among all "permuted test statistics" $t_g\left(\beta_1^0\right)$ for $g \in \mathbf{G}_n$, i.e., $\hat{p} = \frac{1}{M}\sum_{g\in\mathbf{G_n}} \mathbf{1}\left\{t_{\mathrm{id}}\left(\beta_1^0\right) \leq t_g\left(\beta_1^0\right)\right\}$, where $M = |\mathbf{G}_n|$ is the cardinality of $\mathbf{G}_n$.

Toy example. We illustrate tradeoffs involved in the choice of $\mathbf{G}_n$. Consider an instance with $n = 40$ and $p = 2$ with centered outcome variable. Specifically, let $\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \varepsilon$, where $\mathbf{X}_j \in \mathbb{R}^{40}$, $j = 1, 2$.

A naive choice of $\mathbf{G}_n$ would be to use all $40!$ permutations of the observations. However, this would typically produce a set of permuted nuisance regressors $\{g\mathbf{X}_2\}_{g\in\mathbf{G}_n}$ so rich that orthogonalizing $\mathbf{X}_1$ with respect to them in order to produce the residuals $\overline{\mathbf{X}}_1$ would eliminate all explanatory variation, i.e., $\overline{\mathbf{X}}_1 = \mathbf{0}_n$ and $t_g\left(\beta_1^0\right) \equiv 0$, for all g and all β_1^0, leaving the test with no power whatsoever.

We instead break the observations $\{1, 2, \dots, 40\}$ into $n_{\mathrm{blocks}} = 5$ blocks of size eight, $\{1, 2, \cdots, 8\}$, $\{9, \cdots, 16\}$,...,$\{33, \cdots, 40\}$, and consider the $5!$ permutations of those blocks. By Theorem 2 below, the space spanned by all $g\mathbf{X}_2$, $g \in \mathbf{G}_n$, has dimension at most 17, leaving 33 degrees of freedom for explanatory variation in $\overline{\mathbf{X}}_1$. The p-value $\hat{p}$ produced by comparing $t_{\mathrm{id}}\left(\beta_1^0\right)$ to all permuted statistics $t_g\left(\beta_1^0\right)$ changes in increments of $1/(5!)$ as β_1^0 changes, and is thus granular beyond the second digit.

[2]Keeping with the literature, we use the notation $g \in \mathbf{G}_n$ such that its explicit form is clear from context. In particular, $g\mathbf{Z}$ may read as applying a $n \times n$ permutation matrix g to the $n \times 1$ vector $\mathbf{Z}$, and that is equal to $\mathbf{Z}_g = (\mathbf{Z}_{g(1)},\dots,\mathbf{Z}_{g(n)})^T$, where g reads as a bijection on $\{1,\dots,n\}$ corresponding to the same permutation. Specifically, $g : i \mapsto j$ encodes to the matrix operation $g\mathbf{Z} = \mathbf{Z}_g$, where the ith entry of $\mathbf{Z}_g$ is the jth entry of $\mathbf{Z}$.

The requirements for validity compare favorably to those of standard t-tests. While classical t-tests are either exactly valid with Gaussian errors or asymptotically robust when using a robust covariance estimate, but not both, the proposed exact t-test is both asymptotically robust and exact in small samples with merely exchangeable errors. See Table II. This was the explicit motivation of D'Haultfœuille and Tuvaandorj (2024).

Outline. Section 2 develops the proposed test in more detail. Section 3 illustrates the performance of the test in a simulation and an example. Section 4 discusses and concludes. The Supplementary Appendix (Pouliot, 2026) contains additional discussions and technical proofs.

Notation. For a given matrix $\mathbf{Z}$, we denote its column space by $\mathcal{C}(\mathbf{Z})$ and null space by $\mathcal{N}(\mathbf{Z})$. For orthogonal projections, we often use the shorthand $\mathcal{Q}_{\mathbf{Z}} = I - \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^{-}\mathbf{Z}^T$. The annotated equality "$\overset{H_0}{=}$" denotes an equality that holds under the null hypothesis.

2. METHODOLOGY

Consider the proposed statistic (4) with explicit parametrization,

$$t_g\left(\beta_1^0\right) = t\left(\beta_1^0 : \mathbf{X}_1, \mathbf{X}_2, g\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)\right),$$

where we emphasize that the transformation is applied to the "short residuals" $\mathbf{Y} - \mathbf{X}_1\beta_1^0$.

Let $M = |\mathbf{G}_n|$, the cardinality of $\mathbf{G}_n$. Let $\mathbf{W} = (\mathbf{X}_1, \mathbf{X}_2, \mathbf{Y} - \mathbf{X}_1\beta_1^0)$ and use the shorthand $g\mathbf{W} = (\mathbf{X}_1, \mathbf{X}_2, g(\mathbf{Y} - \mathbf{X}_1\beta_1^0))$. For any given $\mathbf{W} \in \operatorname{supp}(\mathbf{W})$, let

$$t^{(1)}(\mathbf{W}) \leq t^{(2)}(\mathbf{W}) \leq \cdots \leq t^{(M)}(\mathbf{W})$$

be the ordered values of $t(\beta_1^0 : g\mathbf{W})$ as g varies over $\mathbf{G}_n$ and write $t^{(i)}$ in lieu of $t^{(i)}(\mathbf{W})$, for any given i, when no confusion arises. Given a nominal level α, let $k = M - \lfloor M\alpha \rfloor$, where $\lfloor M\alpha \rfloor$ denotes the largest integer less than or equal to $M\alpha$.

Let $\hat{t} := t(\beta_1^0 : \mathbf{W})$ be the realized test statistic, and define the test function[3]

$$\phi(\mathbf{W}) = \begin{cases} 1, & \hat{t} > t^{(k)} \\ a, & \hat{t} = t^{(k)} \\ 0, & \hat{t} < t^{(k)} \end{cases},$$

with $a = \left(M\alpha - M^{+}\right)/M^{0}$, where M^{+} and $M^{(0)}$ are the number of values $t^{(j)}$, $j = 1, ..., M$, that are greater than $t^{(k)}$ and equal to $t^{(k)}$, respectively.

2.1. *Exact validity*

Consider the model (2). The *strong null* we are willing to make in order to obtain exact validity is that the errors are exchangeable. We are not, however, willing to assume that the tested and nuisance covariates are exchangeable with respect to each other.

[3]Specifically, when $\hat{t} = t^{(k)}$, the analyst "randomizes" their decision and rejects the null with probability a. In practice, we recommend the slightly conservative but simpler *non-randomized test*

$$\phi(\mathbf{W}) = \begin{cases} 1, & \hat{t} > t^{(k)} \\ 0, & \hat{t} \leq t^{(k)} \end{cases},$$

where $k = \lceil (1-\alpha) M \rceil$. This avoids producing an accept-reject decision that may be random and, as importantly, eliminates an opportunity for p-hacking. See Lei and Sudijono (2024) for a discussion of the latter point.

ASSUMPTION SN—Strong Null: *For any* $g \in \mathbf{G}_n$,

$$(\mathbf{X}_1, \mathbf{X}_2, \boldsymbol{\varepsilon}) \sim (\mathbf{X}_1, \mathbf{X}_2, g\boldsymbol{\varepsilon}) .$$

The symbol "$\sim$" used in $A \sim B$ denotes that A and B have the same distribution. The Strong Null is sufficient for the validity of the proposed test.

THEOREM 1—Validity of Exact t-Test: *Suppose the Strong Null holds and that* $\mathbf{Y} - \mathbf{X}_1\beta_1^0$ *is not in the column span of the constant vector and* $\{g(\mathbf{X}_1, \mathbf{X}_2)\}_{g\in\mathbf{G}_n}$. *Then,*

$$E_{\mathcal{P}}\left[\phi(\mathbf{Y}, \mathbf{X}_1, \mathbf{X}_2)\right] = \alpha. \tag{6}$$

Given the invariance property established in the intuitive construction below, the remainder of the formal proof is standard and is given in Section 1 of the Supplementary Appendix.

Intuition for exact validity. First, consider testing H_0 in the centered univariate case $\mathbf{Y} = \mathbf{X}_1\beta_1 + \varepsilon$. A natural test statistic is $T_g^{\text{simple}}(\beta_1^0) := \mathbf{X}_1^T(g\varepsilon^0)$, where g may for instance vary over the set of all permutations on $\{1, \ldots, n\}$, and where $\varepsilon^0 := \mathbf{Y} - \mathbf{X}_1\beta_1^0$ are the "short residuals". Under the null, $\varepsilon^0 = \varepsilon$ and $T_g^{\text{simple}}(\beta_1^0) \overset{H_0}{=} \mathbf{X}_1^T(g\varepsilon)$, a function of $(\mathbf{X}_1, g\varepsilon)$. The test is thus valid under the strong null that $(\mathbf{X}_1, \varepsilon) \sim (\mathbf{X}_1, g\varepsilon)$, $\forall\, g \in \mathbf{G}_n$.

Now consider the multivariate case $\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \varepsilon$. Under the null hypothesis H_0, the "short residuals" contain the nuisance regressors, i.e., $\varepsilon^0 \overset{H_0}{=} \mathbf{X}_2\beta_2 + \varepsilon$. Consequently, $T_g^{\text{simple}}(\beta_1^0) = \mathbf{X}_1^T(g\varepsilon^0) \overset{H_0}{=} \mathbf{X}_1^T(g\mathbf{X}_2\beta_2) + \mathbf{X}_1^T(g\varepsilon)$ is not invariant with respect to β_2 and need not produce a valid test unless X_{i1} and X_{i2} are independent. Indeed with, say, positively correlated X_{i1} and X_{i2}, $\mathbf{X}_1^T\mathbf{X}_2\beta_2$ will tend to be larger than $\mathbf{X}_1^T(g\mathbf{X}_2\beta_2)$, which will bring about type I error.

Likewise, simply orthogonalizing the short residuals with respect to the nuisance coefficient does not suffice. While the realized statistic $T_{\text{id}}^{\text{orth}}(\beta_1^0) := \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}\varepsilon^0 \overset{H_0}{=} \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}\varepsilon$ is invariant under the null, the statistic $T_g^{\text{orth}}(\beta_1^0) := \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}(g\varepsilon^0) \overset{H_0}{=} \mathbf{X}_1^T\mathcal{Q}_{\mathbf{X}_2}(g\mathbf{X}_2\beta_2 + g\varepsilon)$ is not invariant for $g \neq \text{id}$ since, typically, $\mathcal{Q}_{\mathbf{X}_2}g\mathbf{X}_2\beta_2 \neq 0$. Concretely, $T_{\text{id}}^{\text{orth}}(\beta_1^0)$ would typically not even have the same null marginal resampling distribution as $T_g^{\text{orth}}(\beta_1^0)$ for $g \neq \text{id}$. A longer discussion of intuition via counterexamples is given in the Supplementary Appendix.

Our approach is thus to orthogonalize with respect to $g\mathbf{X}_2$ for every $g \in \mathbf{G}_n$ and propose the –unstudentized– test statistic $T_g(\beta_1^0) = \mathbf{X}_1^T\mathcal{Q}_{\mathbf{GX}_2}g(\mathbf{Y} - \mathbf{X}_1\beta_1^0)$, where $\mathbf{GX}_2 = (g\mathbf{X}_2)_{g\in\mathbf{G}_n} \in \mathbb{R}^{n\times M}$. By construction, $T_g(\beta_1^0) \overset{H_0}{=} \mathbf{X}_1^T\mathcal{Q}_{\mathbf{GX}_2}g\varepsilon$ for any $g \in \mathbf{G}_n$, which is a function of $(\mathbf{X}_1, \mathbf{X}_2, g\varepsilon)$ and thus produces a valid test under the Strong Null.

While the statistic $T_g(\beta_1^0)$ produces an exactly valid test, studentization produces asymptotic robustness without jeopardizing exact validity because the variance estimate is itself a function of $(\mathbf{X}_1, \mathbf{X}_2, g\varepsilon)$ under the null.

A deeper but more abstract intuition also obtains. Consider a general candidate orthogonal projection $\mathcal{Q}$ satisfying $\mathbf{X}_2 \in \mathcal{N}(\mathcal{Q})$. On the one hand, we want a test statistic of the form $\mathbf{X}_1^T g\mathcal{Q}\varepsilon^0$, such that applying $\mathcal{Q}$ to $\varepsilon^0 = \mathbf{X}_2\beta_2 + \varepsilon$ delivers the desired invariance. On the other hand, we want a test statistic of the form $\mathbf{X}_1^T\mathcal{Q}g\varepsilon$ so that the randomization distribution is constructed by directly permuting the residual vector ε, thus delivering exact validity under the Strong Null. This suggests the condition that

$$\mathcal{Q}g = g\mathcal{Q}, \tag{7}$$

as linear operators. Lemma A.1 verifies that $\mathcal{Q}_{\mathbf{GX}_2}$ satisfies the commutativity condition (7).

Remark that validity obtains for any group of actions $\mathbf{G}_n$. It is because of power concerns and for analytical tractability that we focus on subgroups of the group of permutations.

Intuition for power. Writing out

$$T_{\text{id}}\left(\beta_1^0\right) = \overline{\mathbf{X}}_1^T\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right) = \overline{\mathbf{X}}_1^T\mathbf{X}_1\left(\beta_1^0 - \beta_1\right) + \overline{\mathbf{X}}_1^T\varepsilon,$$

where $\overline{\mathbf{X}}_1 = \mathcal{Q}_{\mathbf{GX}_2}\mathbf{X}_1$, reveals that the realized test statistic will better detect departures from the null if $\overline{\mathbf{X}}_1^T\mathbf{X}_1$ is large or, in other words, if enough of the variation in $\mathbf{X}_1$ has been preserved in $\overline{\mathbf{X}}_1$ after projecting onto the orthogonal complement of $\mathcal{C}\left(\mathbf{GX}_2\right)$. However, the matrix $\mathbf{GX}_2$ has as many columns as $\mathbf{G}_n$ has elements, raising *a priori* concerns as to the amount of explanatory variation that must be sacrificed in order to obtain invariance. The important fact making the proposed procedure reasonable is that, if one takes $\mathbf{G}_n$ to be a group of non-overlapping block permutations, then $\mathbf{G}_n$ will have sufficiently large cardinality while maintaining the dimension of $\mathcal{C}\left(\mathbf{GX}_2\right)$, the subspace of foregone explanatory variation, relatively small. This is the content of Theorem 2 and Table I.

Selecting the group of transformations. Indeed, while the procedure is valid in general for a group of actions $\mathbf{G}_n$, the specific choice of $\mathbf{G}_n$ is important.

First, the group of actions $\mathbf{G}_n$ must be large —and thus granular— enough to produce accurate p-values and not forego "too much" statistical power.

Second, as discussed, it must be small enough to leave enough explanatory variation in $\overline{\mathbf{X}}_1 = \mathcal{Q}_{\mathbf{GX}_2}\mathbf{X}_1$ so that the test retains statistical power.

Third, the group of transformations must be simple enough so that a central limit theorem for the randomization distribution can be worked out.

Our suggestion for $\mathbf{G}_n$ is to pick *non-overlapping block permutations*.[4] For instance, for $n = 9$ and number of blocks $n_{\text{blocks}} = 3$, some $g \in \mathbf{G}_n$ gives

$$g\left(1,2,3,\ 4,5,6,\ 7,8,9\right) = \left(4,5,6,\ 7,8,9,\ 1,2,3\right),$$

where the first block is moved to the third block position, the second to the first, and the third to the second.

With respect to our first criterion, the number of permuted test statistics, $n_{\text{blocks}}!$, remains large enough to produce granular randomization distributions even with a small number of blocks. See Table I. Furthermore, Section 4 of the Supplementary Appendix provides evidence that very little power is forgone as a direct consequence of using this smaller group of permutations.

With respect to our second criterion, theory avails which quantifies the dimension of the space that is annihilated by applying the orthogonal projection $\mathcal{Q}_{\mathbf{GX}_2}$ corresponding to $\mathbf{G}_n$.

THEOREM 2: *If* $n \geq n_{\text{blocks}}^2$, *then*

$$\max_{\mathbf{X}\in\mathbb{R}^n} \operatorname{rank}\left(\mathbf{GX}\right) = n_{\text{blocks}} \cdot \left(n_{\text{blocks}} - 2\right) + 2,$$

where $\mathbf{GX} = \left(g^{(1)}\mathbf{X}, \ldots, g^{(M)}\mathbf{X}\right)$.

Remarkably, the block structure makes it such that the rank of $\mathbf{GX}$ increases much slower than M, the number of columns of $\mathbf{GX}$. This is nicely illustrated in Table I.

[4] A permutation $g : \{1,\ldots,n\} \to \{1,\ldots,n\}$ is a *non-overlapping block permutation* if there exist non-intersecting blocks $\mathcal{B}_j \subset \{1,\ldots,n\}$, $j = 1,\ldots,n_{\text{blocks}}$, such that if $i_1, i_2 \in \mathcal{B}_j$, then $g\left(i_1\right), g\left(i_2\right) \in \mathcal{B}_{j'}$ for some j', where i and $g(i)$ have the same position within their respective blocks, and if $i \notin \bigcup_j \mathcal{B}_j$, then $g\left(i\right) = i$.

The proof of Theorem 2 for the general case of a $\mathbf{X} \in \mathbb{R}^{n \times p}$ is given in the Appendix and may be of independent interest as a nice interplay of linear algebra and combinatorics.

Comparison with other methods. Lei and Bickel (2021) and D'Haultfœuille and Tuvaandorj (2024) propose different test statistics and linear action groups which may be used in our framework. This invites a direct conceptual comparison of the methods. See the Supplementary Appendix for discussion of other, less directly related methods.

D'Haultfœuille and Tuvaandorj (2024) produce a test tailored to the case in which the nuisance covariates are discrete and –at least some of– the blocks of observations indexed by the value of the nuisance covariate vector are large.

Their test statistic, unstudentized and in the scalar subvector case, can be expressed as

$$T_g^D\left(\beta_1^0\right) = \tilde{\mathbf{X}}_1^T g\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right), \tag{8}$$

where $\tilde{\mathbf{X}}_1 = \mathcal{Q}^D X_1$, $\mathcal{Q}^D = I - \mathcal{M}^{\mathcal{D}}$ and $\mathcal{M}^{\mathcal{D}}$ is the orthogonal projection onto the column span of block indicator vectors.

They only permute observations for which the nuisance regressors $X_{i,2}$ are equal. This idea dates back to Brown and Maritz (1982). See also Mehra and Sen (1969).

Specifically, D'Haultfœuille and Tuvaandorj (2024) permute within blocks –indexed by the value of $X_{i,2}$– while our proposed procedure permutes across blocks. Because they permute within blocks whose value of the nuisance coefficient $X_{i,2}$ is constant, invariance with respect to $X_{i,2}\beta_2$ is immediate and obtains without orthogonalizing $\mathbf{X}_2$ or any of its permutations. The first-order tradeoff with our method is clear. Unlike with our "orthogonalized" $\overline{\mathbf{X}}_1$, no explanatory variation is forgone by instead using $\tilde{\mathbf{X}}_1$, but the framework is restrictive. In particular, their method does not apply if one of the nuisance regressors is continuous. In situations with all discrete nuisance regressors and with a small number of large blocks indexed by the nuisance covariate vector, our recommendation is to use the procedure of D'Haultfœuille and Tuvaandorj (2024).

Lei and Bickel (2021) produce their test statistic as the solution of a constrained optimization program maximizing a certain proxy for power under constraints enforcing the invariance required for validity.

Heuristically, in lieu of deriving $\overline{\mathbf{X}}_1$ as we did to produce the –unstudentized– statistic $T_g\left(\beta_1^0\right) := \overline{\mathbf{X}}_1^T\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)_g$, they solve for a vector η under invariance and power constraints and propose a statistic of the form $T_g^{LB}\left(\beta_1^0\right) := \eta^T\left(\mathbf{Y} - \mathbf{X}_1\beta_1^0\right)_g$.

Specifically, they require that $\eta^T\left(g\mathbf{X}_2\right)$ be constant for all g in a set of cyclical permutations and that the data generating process satisfy the Strong Null for these permutations.

If we slightly relax their constrained optimization framework, our suggested method may be interpreted as one of its feasible solutions. See the Supplementary Appendix.

One important characterization of the sample size requirement of such methods is the smallest sample size required in order to produce non-randomized p-values that can take value below a pre-specified α, i.e., so that the test can reject at the specified critical level without randomizing the accept-reject decision at ties. Lei and Bickel (2021) produce such a characterization for their test. Their requirement for a non-randomized test with probability of type 1 error at least α is that $n/p \geq 1/\alpha - 1$. Our requirement obtains as theirs do; there must be at least $1/\alpha$ permutations in order to have rejection without randomizing the accept-reject decision. Hence, there must be at least enough blocks such that $n_{\text{blocks}}! \geq 1/\alpha$. For the test to have some power, and thus reject for some alternative, we require $n/p > n_{\text{blocks}}\left(n_{\text{blocks}} - 2\right) + 2$. Consider the case

TABLE I

n_{blocks}	2	3	4	5	6	7	8	9	10
Number of columns	2	6	24	120	720	5,040	40,320	362,880	3,628,800
Rank	2	5	10	17	26	37	50	65	82

Note: Number of columns and maximal rank for a matrix whose columns are $g^{(1)}\mathbf{X}, \ldots, g^{(M)}\mathbf{X}$, allowing a conformable n.

of $\alpha = 0.01$. Lei and Bickel (2021) require $n/p \geq 99$ observations while we require $n/p \geq 18$. And the difference can be made arbitrarily dramatic for smaller values of the critical level α.[5]

2.2. *Large-sample validity*

Our philosophy is that the proposed method has claim to practicality insofar as it is asymptotically valid under assumptions comparable to those required by popular methods, such as the commonly used variants of the t-test. Thus motivated, we produce guarantees of asymptotic validity without the Strong Null, e.g., under conditional heteroskedasticity.

Intuition for asymptotic validity. A randomization test is asymptotically valid when its randomization distribution –one can think of the histogram of $\{t_g(\beta_1^0)\}_{g\in\mathbf{G}_n}$– converges in a proper sense to the null resampling distribution of $t_{\text{id}}(\beta_1^0)$. In that case, even if the randomization test is not exactly valid, the quantiles of the randomization distribution will approximately match the quantiles of the true null resampling distribution of $t_{\text{id}}(\beta_1^0)$, and the test will thus be valid as it indirectly –in some sense, inadvertently– implements the standard large sample frequentist test, which we assume is valid.

The studentization is applied in order to weaken the requirements for asymptotic validity. Considering the unstudentized statistics $T_g(\beta_1^0)$, the histogram of $\{T_g(\beta_1^0)\}_{g\in\mathbf{G}}$ "stabilizes" to take the shape of a $N\left(0, E\left[\bar{X}_{i1}^{*2}\right]E\left[\varepsilon_i^2\right]\right)$, where $\bar{X}_{i1}^*$ is the population analog of $\bar{X}_{i1}$, and the limiting distribution of $T_{\text{id}}(\beta_1^0)$ is $N\left(0, E\left[\bar{X}_{i1}^{*2}\varepsilon_i^2\right]\right)$. These two distributions need not match unless X_{i1} and ε_i are independent, precluding conditional heteroskedasticity. However, for the vast majority of block permutations g, $\hat{\sigma}_g^2$ asymptotically approaches $E\left[\bar{X}_{i1}^{*2}\right]E\left[\varepsilon_i^2\right]$, while $\hat{\sigma}_{\text{id}}^2$ asymptotically approaches $E\left[\bar{X}_{i1}^{*2}\varepsilon_i^2\right]$. Consequently, under weak sufficient conditions that allow for conditional heteroskadasticity, both the histogram of the studentized statistics $\{t_g(\beta_1^0)\}_{g\in\mathbf{G}}$ and the resampling distribution of the unpermuted studentized statistic $t_{\text{id}}(\beta_1^0)$ converge to a $N(0,1)$ distribution. Asymptotic validity thus obtains.

We collect key assumptions.

ASSUMPTIONS M—Moments:
M.1 $E[\varepsilon_i] = 0$ *and* $E[\varepsilon_i X_i] = 0$, $\forall\, i$,
M.2 $E\left[X_{i,j}^4\right], E\left[Y_i^4\right] < \infty$, $\forall\, i, j$.

Assumption M.1 defines the best linear predictor, while M.2 requires bounds on moments which may be considered as standard, see for instance DiCiccio and Romano (2017).

ASSUMPTIONS S—Spectral Conditions: *For* $\overrightarrow{\mathbf{X}} \in \mathbb{R}^{n\times\overrightarrow{p}}$, $\overrightarrow{p} = \text{rank}(\mathbf{G}\mathbf{X}_{-\mathbf{1}})$, *a full rank subset of the columns of* $\mathbf{G}\mathbf{X}_{-\mathbf{1}} = \left(g^{(1)}\mathbf{X}_2, g^{(2)}\mathbf{X}_2, \ldots, g^{(M)}\mathbf{X}_p\right)$, *and* $\check{\mathbf{X}} \in \mathbb{R}^{n\times\check{p}}$, $\check{p} =$

[5] I am grateful to an anonymous referee for this observation.

$\operatorname{rank}(\mathbf{GX})$, *a full rank subset of the columns of* $\mathbf{GX} = \left(g^{(1)}\mathbf{X}_1, ..., g^{(M)}\mathbf{X}_1, g^{(1)}\mathbf{X}_2, \ldots, g^{(M)}\mathbf{X}_p\right)$, *the maximal eigenvalues of the inverse Gram matrices satisfy* $\lambda_{\max}\left(\left(\overrightarrow{\mathbf{X}}^T\overrightarrow{\mathbf{X}}/n\right)^{-1}\right) = O_P\left(\overrightarrow{p}\right)$ *and* $\lambda_{\max}\left(\left(\check{\mathbf{X}}^T\check{\mathbf{X}}/n\right)^{-1}\right) = O_P\left(\check{p}\right)$.

Assumption S is conservative. Indeed, simulations as well as analytical results pertaining to similar cases where different, stronger independence assumptions are imposed (Bai and Yin, 2008, Tao and Vu, 2010) suggest that the largest eigenvalue is bounded in probability. An appreciable feature of this conservative condition is that it can be established under low-level conditions. Lemma SA.2 shows Assumption S holds for continuous data.

ASSUMPTION D—Design: $X_{ij}, \forall\, i, j$, *are uniformly bounded in probability.*

Assumption D restricts the support of the design matrix and, heuristically speaking, requires that arbitrarily large values of the covariates occur with arbitrarily small probability.

We can now produce the theorem stating that even if the strong null does not hold, as long as the *weak null* consisting of the null hypothesis of interest (3) and assumptions including M, S and D hold, then the test ϕ is asymptotically valid.

THEOREM 3: *Suppose* $(Y_i, X_i) \sim \mathcal{P}$ *independently and that Assumptions M, S and D hold. Suppose that* $n_{\text{blocks}}^8/n \to 0$. *Further suppose that* $\varepsilon_i^2 = o_P(n)$, *uniformly, and that* $n_{\text{blocks}}/b^\zeta \to \infty$, *for some* $0 < \zeta < 1$, *where* b *designates the size of blocks. Then, under the null hypothesis* H_0,

$$E_{\mathcal{P}}\left[\phi\left(\mathbf{Y}, \mathbf{X}_1, \mathbf{X}_2\right)\right] \to \alpha,$$

where $\left(\mathbf{Y}, \mathbf{X}_1, \mathbf{X}_2\right) \in \mathbb{R}^{n\times(2+p)}$.

PROOF OF THEOREM 3: Let $J_{t_{\text{id}},n}(\cdot)$ be the sampling distribution of $t_{\text{id}}\left(\beta_1^0\right)$, whose dependence on n is implicit. By Theorem A.2, we have that

$$\lim_{n\to\infty} \sup_{s\in\mathbb{R}} \left|J_{t_{\text{id}},n}(s) - \Phi(s)\right| = 0,$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

Let $\hat{R}_{t_g,n}$ be the permutation distribution of $t_g\left(\beta_1^0\right)$. By Theorem A.4, we have that

$$\hat{R}_{t_g,n}(s) \xrightarrow{P} \Phi(s),$$

for all points of continuity of s.

Invoking Lemma 11.2.1 of Lehmann and Romano (2005), as they do in their Theorem 15.2.3, we find that since $\Phi(\cdot)$ is everywhere strictly increasing and continuous,

$$\hat{R}_{t_g,n}^{-1}(1-\alpha) \xrightarrow{P} \Phi^{-1}(1-\alpha).$$

Because t_{id} is asymptotically normal, it follows by Slutsky's Theorem that

$$\mathrm{E}_{\mathcal{P}}\left[\phi\left(\mathbf{Y}, \mathbf{X}_1, \mathbf{X}_2\right)\right] = \mathcal{P}\left(t_{\text{id}}\left(\beta_1^0\right) > \hat{R}_{t_g,n}^{-1}(1-\alpha)\right) + o(1) \to \mathcal{P}\left(Z > \Phi^{-1}(1-\alpha)\right) = \alpha,$$

where Z denotes a standard normal random variable.

Q.E.D.

The practical implication of Theorem 3 is that by comparing the realized test statistic to the quantiles of the randomization distribution, the randomization test asymptotically rejects according to the standard resampling test, i.e., p-values are effectively computed with respect to the null resampling distribution.

We find that requiring $\varepsilon_i^2 = o_P(n)$ uniformly is not restrictive. It is for instance satisfied and easy to verify in the case of normal random variables. The requirement that $n^8_{\text{blocks}}/n \to 0$ is a direct implication of the conservative rate on the maximal eigenvalue, which is discussed above. The default schedule for the number of blocks used in our statistical package is detailed and discussed in Section 6 of the Supplementary Appendix.

3. SIMULATIONS AND APPLICATIONS

We display some small sample properties of the method. A more exhaustive simulation study and set of applications is presented in Section 4 of the Supplementary Appendix.

TABLE II
COMPARISONS OF VALIDITY

	$n = 25$	$n = 25$	$n = 250$		small n	small n	big n
	Gaussian	Skewed	Hetero.		Gaussian	Skewed	Hetero.
exact t	0.10	0.10	0.10	exact t	yes	yes	yes
classical t	0.10	0.15	0.33	classical t	yes	no	no
robust t	0.14	0.01	0.11	robust t	no	no	yes

Note: Skewed errors are Gamma random variables with shape and rate parameter equal to 0.01 and 1, and rescaled to have variance 1. Heteroskedastic errors are normal random variables with variance $\sigma^2_\varepsilon(X) \propto X_1^2$. The robust t-test is studentized using the robust covariance estimator HC1. The left and right panels report coverage and theoretical validity, respectively. The nominal size is $\alpha = 0.1$.

Table II paints a compelling picture. The classical t-test is exact under Gaussian errors but invalid, even asymptotically, under heteroskedasticity. The robust t-test is invalid in small samples, even with Gaussian errors. The proposed test is valid both in small samples with exchangeable data —e.g., iid Gaussian— and asymptotically under heteroskedasticity.

Table III replicates a regression of college GPA on a measure of high school GPA, a measure of ACT scores, and a measure of the extent to which the student skipped classes. It illustrates the power cost of greater validity. At $n = 141$ and $p = 3$, while the robust t-test may be plausibly valid, small sample robustness may still be desirable. We can see that while the proposed test produces tangibly larger confidence intervals for HS and ACT, it produces a confidence interval of comparable length for skip. See Section 4 of the Supplementary Appendix for more examples and an application with instrumental variables.

4. CONCLUSION

Regression analysts seeking robust inference may find the proposed method useful; while the standard t-test toolkit requires to choose between small sample validity and asymptotic robustness, the proposed test delivers both.

Regression analysts implementing "placebo" or "falsification randomization" tests of a regressor by replacing it with permuted versions of itself may likewise find the procedure useful. Contrasting with and complementing Young (2019), who corroborates the claim that experimental data should often be analyzed using a Fisher test, we emphasize that permutation tests

TABLE III

REANALYZING EXAMPLE 4.3 OF WOOLDRIDGE (2020).

	$\hat{\beta}$	exact t	p-val	classical t	p-val	robust t	p-val
HS	0.412	(0.252, 0.701)	0.017	(0.257, 0.567)	0.000	(0.249, 0.574)	0.000
ACT	0.015	(-0.013, 0.035)	0.342	(-0.003, 0.032)	0.166	(-0.004, 0.033)	0.184
skip	-0.083	(-0.135, -0.047)	0.008	(-0.126, -0.040)	0.002	(-0.126, -0.040)	0.002

Note: Critical level is $\alpha = 0.1$, $n = 141$ and $n_{\text{blocks}} = 5$. The robust t-test uses variance estimate HC1.

on observational data are rarely conditionally valid as Fisher tests, but are often plausibly unconditionally valid as exact randomization tests. However, as detailed herein, naively implemented permutation tests for regression coefficients are typically invalid in small samples. The proposed test then offers a valid alternative.

APPENDIX A: MATHEMATICAL APPENDIX

LEMMA A.1: *Let $\mathbf{G}_n$ be a finite group of orthogonal linear transformations of $\mathbb{R}^n$ onto itself. Let $\mathbf{Z} = (\mathbf{Z}_1, \ldots, \mathbf{Z}_p) \in \mathbb{R}^{n \times p}$ be a given matrix. Let $\mathcal{Q}$ be the orthogonal projection onto $\mathcal{C}^{\perp}\left(g^{(1)}\mathbf{Z}_1, \ldots, g^{(M)}\mathbf{Z}_1, g^{(1)}\mathbf{Z}_2, \ldots, g^{(M)}\mathbf{Z}_p\right)$. Then, for any $\mathbf{X} \in \mathbb{R}^n$,*

$$g\mathcal{Q}\mathbf{X} = \mathcal{Q}g\mathbf{X}, \ \forall\, g \in \mathbf{G}_n. \tag{9}$$

PROOF OF LEMMA A.1: Denote, for short, the column space $\mathcal{C}(\mathbf{GZ}) = \mathcal{C}\left(g^{(1)}\mathbf{Z}_1, \ldots, g^{(M)}\mathbf{Z}_1, g^{(1)}\mathbf{Z}_2, \ldots, g^{(M)}\mathbf{Z}_p\right)$ and its orthogonal complement $\mathcal{C}^{\perp}(\mathbf{GZ})$. Observe that $\mathcal{Q} = I - \mathcal{M}$, where $\mathcal{M}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{GZ})$. Because (9) holds if and only if $g\mathcal{M}\mathbf{X} = \mathcal{M}g\mathbf{X}$, it suffices to show the latter.

For any $\mathbf{X} \in \mathbb{R}^n$, we may produce the decomposition

$$\mathbf{X} = u + v, \ u \in \mathcal{C}(\mathbf{GZ}), \ v \in \mathcal{C}^{\perp}(\mathbf{GZ}).$$

We collect two observations. First, observe that $u \in \mathcal{C}(\mathbf{GZ})$ implies that $u = \sum_{j=1}^{p}\sum_{i=1}^{M} a_{i,j} g^{(i)} Z_j$, for some real numbers $a_{i,j}$, $j = 1, \ldots, p$, $i = 1, \ldots, M$. Hence, for any $g \in \mathbf{G}_n$,

$$gu = g\sum_{j=1}^{p}\sum_{i=1}^{M} a_{i,j} g^{(i)} Z_j = \sum_{j=1}^{p}\sum_{i=1}^{M} a_{i,j} \underbrace{g g^{(i)}}_{\in \mathbf{G}_n} Z_j \in \mathcal{C}(\mathbf{GZ}).$$

Second, observe that $v \in \mathcal{C}^{\perp}(\mathbf{GZ})$ can (for a general v) equivalently be stated as $v \perp g^{(1)}\mathbf{Z}_1, \ldots, g^{(M)}\mathbf{Z}_1, g^{(1)}\mathbf{Z}_2, \ldots, g^{(M)}\mathbf{Z}_p$. Hence,

$$(gv)^T\left(g^{(i)}\mathbf{Z}_j\right) = v^T\left(\underbrace{g^{-1}g^{(i)}}_{\in \mathbf{G}_n}\mathbf{Z}_j\right) = 0,$$

for any i and j. Which is to say, $gv \in \mathcal{C}^{\perp}(\mathbf{GZ})$.

This means that

$$g\mathcal{M}\mathbf{X} = g\mathcal{M}(u + v) = g(\mathcal{M}u + \mathcal{M}v) = g\mathcal{M}u = gu,$$

and

$$\mathcal{M}g\mathbf{X} = \mathcal{M}g\left(u+v\right) = \mathcal{M}gu + \mathcal{M}gv \overset{(a)}{=} \mathcal{M}gu \overset{(b)}{=} gu,$$

where (a) holds because of the second observation, and (b) holds because of the first observation. *Q.E.D.*

THEOREM A.1—General Case: *If* $n \geq p \cdot n_{\text{blocks}}^2$, *then*

$$\max_{\mathbf{X}\in\mathbb{R}^{n\times p}} \operatorname{rank}\left(\mathbf{GX}\right) = p \cdot \left(n_{\text{blocks}} \cdot \left(n_{\text{blocks}} - 2\right) + 2\right),$$

where $\mathbf{GX} = \left(g^{(1)}\mathbf{X}, \ldots, g^{(M)}\mathbf{X}\right)$.

We present the proof of Theorem A.1 in two parts. As a Lemma, and to simplify the exposition of the general case, we first prove the case with $p = 1$.

PROOF OF THEOREM A.1, THE $p = 1$ CASE: We can provide a proof by induction. For any admissible n and n_{blocks}, designate the blocks of the n-tuple $\mathbf{X}$ by $A, B, \cdots \in \mathbb{R}^b$, where $A = \left(X_1, \ldots, X_b\right)^T$, $B = \left(X_{b+1}, \ldots, X_{2b}\right)^T$, and so on.

In the base case with $n_{\text{blocks}} = 2$, we have

$$\mathbf{G}_{b\cdot 2}\mathbf{X} = \left(\begin{pmatrix} A \\ B \end{pmatrix}, \begin{pmatrix} B \\ A \end{pmatrix}\right),$$

which obviously has $\max_{\mathbf{X}\in\mathbb{R}^n} \operatorname{rank}\left(\mathbf{G}_{b\cdot 2}\mathbf{X}\right) = 2$.

Now consider any given number of blocks n_{blocks}. Consider splitting all possible block permutations into sets according to their first block. Specifically, consider

$$\underbrace{\begin{pmatrix} A \\ B \\ \vdots \\ Y \\ Z \end{pmatrix}, \begin{pmatrix} A \\ B \\ \vdots \\ Z \\ Y \end{pmatrix}, \ldots, \begin{pmatrix} A \\ Z \\ \vdots \\ \vdots \\ B \end{pmatrix}}_{\mathcal{S}_1}, \underbrace{\begin{pmatrix} B \\ A \\ \vdots \\ Y \\ Z \end{pmatrix}, \begin{pmatrix} B \\ A \\ \vdots \\ Z \\ Y \end{pmatrix}, \ldots, \begin{pmatrix} B \\ Z \\ \vdots \\ \vdots \\ A \end{pmatrix}}_{\mathcal{S}_2}, \ldots, \underbrace{\begin{pmatrix} Z \\ A \\ \vdots \\ W \\ Y \end{pmatrix}, \begin{pmatrix} Z \\ A \\ \vdots \\ Y \\ W \end{pmatrix}, \ldots, \begin{pmatrix} Z \\ Y \\ \vdots \\ \vdots \\ A \end{pmatrix}}_{\mathcal{S}_{n_{\text{blocks}}}},$$

where the set $\mathcal{S}_1$ has block A in the first block of rows, and lists all permutations of the other blocks $B, \ldots, Z$ over all its vectors. The set $\mathcal{S}_2$ has block B in its first block of rows and lists all permutations of the other blocks $A, C, \ldots, Z$ over all its vectors, and so on. To be sure, in this lexicographic order, Z is the $n_{\text{blocks}}^{\text{th}}$ block, not necessarily the 26th block.

For exposition purposes, consider the following reordering of the columns of $\mathbf{GX}$,

$$\underbrace{\begin{pmatrix} A \\ B \\ \vdots \\ Y \\ Z \end{pmatrix}, \begin{pmatrix} A \\ B \\ \vdots \\ Z \\ Y \end{pmatrix}, \ldots, \begin{pmatrix} A \\ Z \\ \vdots \\ \vdots \\ B \end{pmatrix}}_{\mathcal{S}_1}, \underbrace{\begin{pmatrix} B \\ A \\ \vdots \\ Y \\ Z \end{pmatrix}, \begin{pmatrix} B \\ C \\ A \\ \vdots \\ \vdots \end{pmatrix}, \ldots, \begin{pmatrix} B \\ C \\ \vdots \\ \vdots \\ A \end{pmatrix}, \ldots, \begin{pmatrix} B \\ Y \\ \vdots \\ \vdots \\ C \end{pmatrix}}_{\mathcal{S}_2}, \ldots, \underbrace{\begin{pmatrix} Z \\ A \\ \vdots \\ W \\ Y \end{pmatrix}, \begin{pmatrix} Z \\ A \\ \vdots \\ Y \\ W \end{pmatrix}, \ldots, \begin{pmatrix} Z \\ Y \\ \vdots \\ \vdots \\ A \end{pmatrix}}_{\mathcal{S}_{n_{\text{blocks}}}},$$

where we reordered the columns of the set $\mathcal{S}_2$ to emphasize that we can have the first $n_{\text{blocks}} - 1$ vectors of $\mathcal{S}_2$ with block A in the second, third,..., and $n_{\text{blocks}}^{\text{th}}$ block of rows, respectively.

By inductive hypothesis, the rank of $\mathcal{S}_1$ is at least $(n_{\text{blocks}}-1)\cdot(n_{\text{blocks}}-3)+2$. Now the key is that, since $n \geq n_{\text{blocks}}^2$, which is to say, $b \geq n_{\text{blocks}}$, we can always pick an $\mathbf{X}$ such the block A cannot be produced as a linear combination of the $n_{\text{blocks}}-1$ blocks $B, C \ldots, Z$. Appending one-by-one to $\mathcal{S}_1$ vectors for which A shows up in a block of rows for the first time, we must each time increase the rank by 1. Hence, appending S_2 —and thus its first $n_{\text{blocks}}-1$ vectors in the reordering above— to $\mathcal{S}_1$, we must increase the rank by at least $n_{\text{blocks}}-1$. By the same logic, considering the first block of rows, appending one-by-one to $(\mathcal{S}_1, \mathcal{S}_2)$ the sets $\mathcal{S}_3, \ldots, \mathcal{S}_{n_{\text{blocks}}}$ must increase the rank by at least 1 each time. Therefore,

$$\begin{aligned}\max_{\mathbf{X}\in\mathbb{R}^n} \operatorname{rank}(\mathbf{GX}) &\geq (n_{\text{blocks}}-1)\cdot(n_{\text{blocks}}-3)+2+n_{\text{blocks}}-1+n_{\text{blocks}}-2\\ &= n_{\text{blocks}}\cdot(n_{\text{blocks}}-2)+2.\end{aligned}$$

To complete the argument, it suffices to show that any column other than the ones invoked above to lower bound the rank —in fact, any column— can be produced as a linear combination of the said used columns. Specifically, it suffices to show that

$$\underbrace{\begin{pmatrix}A\\B\\\vdots\\Y\\Z\end{pmatrix}, \begin{pmatrix}A\\B\\\vdots\\Z\\Y\end{pmatrix}, \ldots, \begin{pmatrix}A\\Z\\\vdots\\\vdots\\B\end{pmatrix}}_{\mathcal{S}_1}, \underbrace{\begin{pmatrix}B\\A\\\vdots\\Y\\Z\end{pmatrix}, \begin{pmatrix}B\\C\\A\\\vdots\\\vdots\end{pmatrix}, \ldots, \begin{pmatrix}B\\C\\\vdots\\\vdots\\A\end{pmatrix}}_{n_{\text{blocks}}-1 \text{ vectors from } \mathcal{S}_2}, \underbrace{\begin{pmatrix}C\\A\\B\\\vdots\\Z\end{pmatrix}}_{1 \text{ vector from } \mathcal{S}_3} \ldots, \underbrace{\begin{pmatrix}Z\\A\\B\\\vdots\\Z\end{pmatrix}}_{1 \text{ vector from } \mathcal{S}_{n_{\text{blocks}}}},$$

span all columns of $\mathbf{GX}$. Note that the columns of $\mathcal{S}_1$ are not linearly independent —we could instead extract the first $(n_{\text{blocks}}-1)\cdot(n_{\text{blocks}}-3)$ columns in this ordering, they are linearly independent— but we know the rank of the set.[6]

As a first step, we show that all columns in the second set $\mathcal{S}_2$ are in the span of $\mathcal{S}_1$ and the $n_{\text{blocks}}-1$ vectors selected from $\mathcal{S}_2$. Pick any vector from $\mathcal{S}_2$; it has block B in the first block of rows. Without loss of generality, say A is in the second block of entries. I.e., the vector has form

$$\begin{pmatrix}B^T & A^T & A_3^T & \cdots & A_{n_{\text{blocks}}}^T\end{pmatrix}^T, \tag{10}$$

for some $A_3, \ldots, A_{n_{\text{blocks}}}$. In the vectors selected from $\mathcal{S}_2$, we have

$$\begin{pmatrix}B^T & A^T & A_3'^T & \cdots & A_{n_{\text{blocks}}}'^T\end{pmatrix}^T, \tag{11}$$

for some $A_3', \ldots, A_{n_{\text{blocks}}}'$. Remark that $\{A_3, \ldots, A_{n_{\text{blocks}}}\} = \{A_3', \ldots, A_{n_{\text{blocks}}}'\} = \{C, \ldots, Z\}$. Further remark that, in $\mathcal{S}_1$, we have the vectors

$$\begin{pmatrix}A^T & B^T & A_3^T & \cdots & A_{n_{\text{blocks}}}^T\end{pmatrix}^T, \begin{pmatrix}A^T & B^T & A_3'^T & \cdots & A_{n_{\text{blocks}}}'^T\end{pmatrix}^T,$$

[6]As pointed out by an anonymous referee, one may immediately see why the lower bound of $(n_{\text{blocks}}-1)\cdot(n_{\text{blocks}}-3)$ on $\operatorname{rank}(\mathcal{S}_1)$ is attained and not exceeded. Let $A\mathbf{1}^T$ be the submatrix of $\mathcal{S}_1$ containing its b first rows. Define M as the lower submatrix containing the remaining rows. The induction hypothesis applies to M. However, all columns of M have the same sum, i.e., $m\cdot\mathbf{1}^T = \mathbf{1}^T M$ with m the sum of all elements in $B, C, \ldots Z$. If $m \neq 0$, vertically appending a matrix $A\mathbf{1}^T$ whose rows are themselves constant, and thus in the row span of M, cannot increase the (row) rank. If $m = 0$, you can replace M by $M + \epsilon\cdot\mathbf{1}\mathbf{1}^T$ without affecting the rank of $\mathcal{S}_1$ since the modification obtains by adding scalar multiples of a row of $A\mathbf{1}^T$. Then repeat the argument for $m \neq 0$.

which span the difference

$$\begin{pmatrix} 0_b^T & 0_b^T & (A_3 - A_3')^T & \cdots & (A_{n_{\text{blocks}}} - A'_{n_{\text{blocks}}})^T \end{pmatrix}^T.$$

Adding this vector to (11) produces (10).

Now consider any column from the third to $n_{\text{blocks}}^{\text{th}}$ set. Without loss of generality, take it from the third set $\mathcal{S}_3$. The vector has the form

$$\begin{pmatrix} C^T & A_2^T & A_3^T & \cdots & A_{n_{\text{blocks}}}^T \end{pmatrix}^T \tag{12}$$

for some $A_2, A_3, \ldots, A_{n_{\text{blocks}}}$. In the candidate spanning set, we have the vector

$$\begin{pmatrix} C^T & A_2'^T & A_3'^T & \cdots & A'^T_{n_{\text{blocks}}} \end{pmatrix}^T \tag{13}$$

for some $A_2', A_3', \ldots, A'_{n_{\text{blocks}}}$. Remark that $\{A_2, A_3, \ldots, A_{n_{\text{blocks}}}\} = \{A_2', A_3', \ldots, A'_{n_{\text{blocks}}}\} = \{B, D, \ldots, Z\}$. Without loss of generality, let $A_2 = A$ and $A_3 = B$. I.e., we are trying to produce

$$\begin{pmatrix} C^T & A^T & B^T & \cdots & A_{n_{\text{blocks}}}^T \end{pmatrix}^T \tag{14}$$

form the candidate spanning set.

We need to consider cases. First, consider the case in which $A_2' = A$. We can produce

$$\begin{pmatrix} 0^T & 0^T & (B - A_3')^T & \cdots & (A_{n_{\text{blocks}}} - A'_{n_{\text{blocks}}})^T \end{pmatrix}^T$$

from set $\mathcal{S}_1$, add it to (13), and produce (14).

Second, consider the case in which $A_3' = A$. From set $\mathcal{S}_1$, you can produce the vector

$$\begin{pmatrix} 0^T & (C - A_2')^T & (B - C)^T & (A_4 - A_4')^T & \cdots & \cdots & (A_{n_{\text{blocks}}} - A'_{n_{\text{blocks}}})^T \end{pmatrix}^T. \tag{15}$$

From set $\mathcal{S}_2$, you can produce the vector

$$\begin{pmatrix} 0^T & (A - C)^T & (C - A)^T & \cdots & 0'^T \end{pmatrix}^T. \tag{16}$$

Adding (15) and (16) to (13), we produce (14).

Finally, consider the third case in which $A_j' = A$, $j \in \{4, \ldots, n_{\text{blocks}}\}$. Without loss of generality, suppose $A_4' = A$. From set $\mathcal{S}_1$, you can produce the vector

$$\begin{pmatrix} 0^T & (C - A_2')^T & (B - A_3')^T & (A_4 - C)^T & (A_5 - A_5')^T & \cdots & (A_{n_{\text{blokcs}}} - A'_{n_{\text{blokcs}}})^T \end{pmatrix}^T. \tag{17}$$

From the set $\mathcal{S}_2$, you can produce the vector

$$\begin{aligned} &\begin{pmatrix} 0^T & (A - C)^T & 0^T & (C - A_4')^T & 0^T & \cdots & 0^T \end{pmatrix}^T \\ =\ &\begin{pmatrix} 0^T & (A - C)^T & 0^T & (C - A)^T & 0^T & \cdots & 0^T \end{pmatrix}^T. \end{aligned} \tag{18}$$

Adding (17) and (18) to (13), we produce (14).

Q.E.D.

PROOF OF THEOREM A.1, THE $p < n$ CASE: We can provide a proof by induction for any admissible n and n_{blocks}. For each $j = 1, \ldots, p$, designate the blocks of the n-tuple $\mathbf{X}_j$ by $A_j, B_j, \cdots \in \mathbb{R}^b$, where $A_j = (X_{1,j}, \ldots, X_{b,j})^T$, $B_j = (X_{b+1,j}, \ldots, X_{2b,j})$, and so on.

In the base case with $n_{\text{blocks}} = 2$, we have

$$\mathbf{G}_{b\cdot 2}\mathbf{X} = \left(\begin{pmatrix} A_1 \\ B_1 \end{pmatrix}, \begin{pmatrix} B_1 \\ A_1 \end{pmatrix}, \begin{pmatrix} A_2 \\ B_2 \end{pmatrix}, \cdots, \begin{pmatrix} B_p \\ A_p \end{pmatrix} \right),$$

which obviously has $\max_{\mathbf{X}\in\mathbb{R}^n} \operatorname{rank}(\mathbf{G}_{b\cdot 2}\mathbf{X}) = 2p$.

Now consider any given number of blocks n_{blocks}. For each $j = 1, \ldots, p$, consider splitting all possible block permutations of $\mathbf{X}_j$ into sets according to their first block. Specifically, consider

$$\underbrace{\begin{pmatrix} A_j \\ B_j \\ \vdots \\ Y_j \\ Z_j \end{pmatrix}, \begin{pmatrix} A_j \\ B_j \\ \vdots \\ Z_j \\ Y_j \end{pmatrix}, \ldots, \begin{pmatrix} A_j \\ Z_j \\ \vdots \\ \vdots \\ B_j \end{pmatrix}}_{\mathcal{S}_{1,j}}, \underbrace{\begin{pmatrix} B_j \\ A_j \\ \vdots \\ Y_j \\ Z_j \end{pmatrix}, \begin{pmatrix} B_j \\ A_j \\ \vdots \\ Z_j \\ Y_j \end{pmatrix}, \ldots, \begin{pmatrix} B_j \\ Z_j \\ \vdots \\ \vdots \\ A_j \end{pmatrix}}_{\mathcal{S}_{2,j}}, \ldots, \underbrace{\begin{pmatrix} Z_j \\ A_j \\ \vdots \\ W_j \\ Y_j \end{pmatrix}, \begin{pmatrix} Z_j \\ A_j \\ \vdots \\ Y_j \\ W_j \end{pmatrix}, \ldots, \begin{pmatrix} Z_j \\ Y_j \\ \vdots \\ \vdots \\ A_j \end{pmatrix}}_{\mathcal{S}_{n_{\text{blocks}},j}},$$

where the set $\mathcal{S}_{1,j}$ has block A_j in the first block of rows, and lists all permutations of the other blocks $B_j, \ldots, Z_j$ over all its vector. The set $\mathcal{S}_{2,j}$ has block B_j in its first block of rows and lists all permutations of the other blocks $A_j, C_j, \ldots, Z_j$ over all its vectors, and so on. To be sure, Z_j is the $n_{\text{blocks}}^{\text{th}}$ block, not necessarily the 26th block.

For exposition purposes, consider the following reordering of the columns of $\mathbf{G}\mathbf{X}_j$, for each $j = 1, \ldots, p$, as

$$\underbrace{\begin{pmatrix} A_j \\ B_j \\ \vdots \\ Y_j \\ Z_j \end{pmatrix}, \begin{pmatrix} A_j \\ B_j \\ \vdots \\ Z_j \\ Y_j \end{pmatrix}, \ldots, \begin{pmatrix} A_j \\ Z_j \\ \vdots \\ \vdots \\ B_j \end{pmatrix}}_{\mathcal{S}_{1,j}}, \underbrace{\begin{pmatrix} B_j \\ A_j \\ \vdots \\ Y \\ Z \end{pmatrix}, \begin{pmatrix} B_j \\ C_j \\ A_j \\ \vdots \\ \vdots \end{pmatrix}, \ldots, \begin{pmatrix} B_j \\ C_j \\ \vdots \\ \vdots \\ A_j \end{pmatrix}, \ldots, \begin{pmatrix} B_j \\ Y_j \\ \vdots \\ \vdots \\ C_j \end{pmatrix}}_{\mathcal{S}_{2,j}}, \ldots, \underbrace{\begin{pmatrix} Z_j \\ A_j \\ \vdots \\ W_j \\ Y_j \end{pmatrix}, \begin{pmatrix} Z_j \\ A_j \\ \vdots \\ Y_j \\ W_j \end{pmatrix}, \ldots, \begin{pmatrix} Z_j \\ Y_j \\ \vdots \\ \vdots \\ A_j \end{pmatrix}}_{\mathcal{S}_{n_{\text{blocks}},j}},$$

where we reordered the columns of each set $\mathcal{S}_{2,j}$ to emphasize that we can have the first $n_{\text{blocks}} - 1$ vectors of each $\mathcal{S}_{2,j}$ with block A_j in the second, third,..., and $n_{\text{blocks}}^{\text{th}}$ block of rows, respectively.

Consider

$$\mathcal{S}_1 := \mathcal{S}_{1,1} \cup \mathcal{S}_{1,2} \cup \cdots \cup \mathcal{S}_{1,p}.$$

Because $n \geq n_{\text{blocks}}^2 \cdot p$, we have that $b \geq n_{\text{blocks}} \cdot p$ and thus, for any j, A_j does not obtain as a linear combination of vectors in $\{A_l\}_{l\neq j} \bigcup \{B_j, C_j, \ldots\}_{j=1}^{p}$ and thus, invoking the induction hypothesis, we see that

$$\begin{aligned} \operatorname{rank}(\mathcal{S}_1) &= \operatorname{rank}(\mathcal{S}_{1,1}) + \operatorname{rank}(\mathcal{S}_{1,2}) + \cdots + \operatorname{rank}(\mathcal{S}_{1,p}) \\ &= p \cdot ((n_{\text{blocks}} - 1) \cdot (n_{\text{blocks}} - 3) + 2). \end{aligned}$$

In the argument below, we invoke vectors and subvectors of $\mathbf{X}$ being linearly independent if the cardinality of the set they make up is less than the dimensionality of the vectors or subvectors. Repeatedly invoking this should not raise concerns about existence; an $\mathbf{X}$ whose entries are drawn from a distribution that is uniformly continuous with respect to Lebesgue measure will satisfy all of these this with probability one.

Now consider adding the first column of the reordered set $\mathcal{S}_{2,1}$. A_1 does not appear in the second block of rows of any columns of $\mathcal{S}_1$ and A_1 cannot be produced as a linear combination

of the second block of rows for any columns in $\mathcal{S}_1$ since there are $p \cdot (n_{\text{blocks}} - 1)$ distinct blocks in the second block of rows, and $b > p \cdot (n_{\text{blocks}} - 1)$. Appending the first column of the reordered set $\mathcal{S}_{2,1}$ must this increase the maximum rank by 1.

Let $\mathcal{S}$ denote $\mathcal{S}_1$ appended with the first column of the reordered set $\mathcal{S}_{2,1}$. Now consider adding the second column of the reordered set $\mathcal{S}_{2,1}$ to $\mathcal{S}$. A_1 does not appear in the third block of rows of any columns of $\mathcal{S}$ with and A_1 cannot be produced as a linear combination of the second block of rows for any columns in $\mathcal{S}_1$ since there are $p \cdot (n_{\text{blocks}} - 1)$ distinct blocks in the second block of rows, and $b > p \cdot (n_{\text{blocks}} - 1)$.

The same argument holds analogously for the 3rd to $n_{\text{blocks}}^{\text{th}}$ vector of the reordered set $\mathcal{S}_{2,1}$. Hence, appending $S_{2,1}$ to $\mathcal{S}_1$ must increase the rank by $n_{\text{blocks}} - 1$. In fact, the same argument holds for the first —or any— vector of $S_{3,1}$,...,$S_{n_{\text{blocks}},1}$. Hence, appending $S_{3,1} \cup \cdots \cup S_{n_{\text{blocks}},1}$ to $\mathcal{S}$ must increase the rank by $n_{\text{blocks}} - 2$. This establishes the lower bound

$$\begin{aligned}\text{rank}\,(\mathbf{GX}) &\geq p \cdot ((n_{\text{blocks}} - 1) \cdot (n_{\text{blocks}} - 3) + 2) + p \cdot (n_{\text{blocks}} - 1 + n_{\text{blocks}} - 2) \\ &= p \cdot (n_{\text{blocks}} \cdot (n_{\text{blocks}} - 2) + 2)\,.\end{aligned}$$

To see that the bound may be attained, apply the argument of the $p = 1$ proof to show that within each subset composed of $\mathcal{S}_{i_{\text{blocks}},j}$, $i_{\text{blocks}} = 1, \ldots, n_{\text{blocks}}$, we can generate the full set from $n_{\text{blocks}} \cdot (n_{\text{blocks}} - 2) + 2$ linearly independent column vectors identified above to lower bound its rank. *Q.E.D.*

A.1. *Large Sample Theory*

The model is

$$\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \cdots + \mathbf{X}_p\beta_p + \boldsymbol{\varepsilon},$$

where $\mathbf{1}_n \in \mathcal{C}\,(\mathbf{X}_2, \ldots, \mathbf{X}_p)$. Let $\overline{\mathbf{X}}_1 = \mathcal{Q}_{\mathbf{GX}_{-1}}\mathbf{X}_1$, where $\mathbf{GX}_{-1} = \{g\mathbf{X}_2, \ldots, g\mathbf{X}_p\}_{g\in\mathbf{G}_n}$, and the residuals $\hat{\varepsilon}_i$ obtain from the regression of $\mathbf{Y}$ on $\{g\mathbf{X}_1\}_{g\in\mathbf{G}_n}$ and $\mathbf{GX}_{-1}$.

For the resampling asymptotic theory, we are interested in the statistic $t_{\text{id}}\,(\beta_1^0) = n^{-1/2}\mathbf{X}_1^T\mathcal{Q}\,(\mathbf{Y} - \mathbf{X}_1\beta_1^0)$ where $\hat{\sigma}_{\text{id}}^2 = \frac{1}{n}\sum_{i=1}^n \overline{\mathbf{X}}_{i1}^2\hat{\varepsilon}_i^2$.

THEOREM A.2: *Let $J_{t_{\text{id}},n}\,(\cdot)$ be the sampling distribution of $t_{\text{id}}\,(\beta_1^0)$. Suppose $(Y_i, X_i) \sim \mathcal{P}$ independently, and that Assumptions M, S and D hold. Suppose that $n_{\text{blocks}}^8/n \to 0$. Then, under the null hypothesis (3),*

$$\lim_{n\to\infty} \sup_{s\in\mathbb{R}} |J_{t_{\text{id}},n}\,(s) - \Phi\,(s)| = 0,$$

where $\Phi\,(\cdot)$ is the standard normal cumulative distribution function.

PROOF OF THEOREM A.2: Let $\bar{X}_{i1}^*$ be the population analog of $\bar{X}_{i1}$. The numerator of (4) satisfies

$$\frac{1}{\sqrt{n}}\overline{\mathbf{X}}_1^T\boldsymbol{\varepsilon} \xrightarrow{d} N\left(0, E\left(\bar{X}_{i1}^{*2}\varepsilon_i^2\right)\right). \tag{19}$$

To see this explicitly, construct a full rank submatrix $\overrightarrow{\mathbf{X}} \in \mathbb{R}^{n\times\overrightarrow{p}}$ satisfying $\mathcal{C}\left(\overrightarrow{\mathbf{X}}\right) = \mathcal{C}\,(\mathbf{GX}_{-1})$ and Assumption S.

Assume without loss of generality that the maximum rank is attained and $\overrightarrow{p} = (p-1)\,(n_{\text{blocks}} \cdot (n_{\text{blocks}} -$

Let $\overline{\mathbf{X}}_1 = (I - \mathbf{M})\,\mathbf{X}_1 =: \mathbf{X}_1 - \overrightarrow{\mathbf{X}}\hat{\gamma}$, where $\mathbf{M}$ is the orthogonal projection onto $\mathcal{C}\left(\overrightarrow{\mathbf{X}}\right)$. Let $\overline{\mathbf{X}}_1^* = \mathbf{X}_1 - E^*\left[\mathbf{X}_1 \middle| \overrightarrow{\mathbf{X}}\right]$, where $E^*\left[\mathbf{X}_1 \middle| \overrightarrow{\mathbf{X}}\right] =: \overrightarrow{\mathbf{X}}\gamma$ stands for the best linear predictor of $\mathbf{X}_1$ given $\overrightarrow{\mathbf{X}}$.[7] Note that the numerator of (4) satisfies

$$\frac{1}{\sqrt{n}}\overline{\mathbf{X}}_1^T\boldsymbol{\varepsilon} = \frac{1}{\sqrt{n}}\overline{\mathbf{X}}_1^{*T}\boldsymbol{\varepsilon} - \frac{1}{\sqrt{n}}\overline{\mathbf{X}}_1^{*T}\mathbf{M}\boldsymbol{\varepsilon}.$$

The first term satisfies

$$\frac{1}{\sqrt{n}}\overline{\mathbf{X}}_1^{*T}\boldsymbol{\varepsilon} \xrightarrow{d} N\left(0, E\left(\bar{X}_1^{*2}\varepsilon^2\right)\right), \tag{20}$$

by Assumption M.

The second term has order of magnitude at most $O_P\left(\overrightarrow{p}^2/\sqrt{n}\right)$ because

$$\frac{1}{\sqrt{n}}\left|\frac{\overline{\mathbf{X}}_1^{*T}\overrightarrow{\mathbf{X}}}{\sqrt{n}}\left(\frac{\overrightarrow{\mathbf{X}}^T\overrightarrow{\mathbf{X}}}{n}\right)^{-1}\frac{\overrightarrow{\mathbf{X}}^T\boldsymbol{\varepsilon}}{\sqrt{n}}\right| \leq \frac{1}{\sqrt{n}}\left\|\frac{\overline{\mathbf{X}}_1^{*T}\overrightarrow{\mathbf{X}}}{\sqrt{n}}\right\|_2 \lambda_{\max}\left(\left(\frac{\overrightarrow{\mathbf{X}}^T\overrightarrow{\mathbf{X}}}{n}\right)^{-1}\right)\left\|\frac{\overrightarrow{\mathbf{X}}^T\boldsymbol{\varepsilon}}{\sqrt{n}}\right\|_2, \tag{21}$$

where $\lambda_{\max}\left(\left(\frac{\overrightarrow{\mathbf{X}}^T\overrightarrow{\mathbf{X}}}{n}\right)^{-1}\right) = O_P\left(\overrightarrow{p}\right)$, $\left\|\overline{\mathbf{X}}_1^{*T}\overrightarrow{\mathbf{X}}/\sqrt{n}\right\|_2 = O_P\left(\sqrt{\overrightarrow{p}}\right)$, and $\left\|\overrightarrow{\mathbf{X}}^T\boldsymbol{\varepsilon}/\sqrt{n}\right\|_2 = O_P\left(\sqrt{\overrightarrow{p}}\right)$.

The rate on the norm $\left\|\overline{\mathbf{X}}_1^{*T}\overrightarrow{\mathbf{X}}/\sqrt{n}\right\|_2$ obtains by Markov inequality because

$$P\left(\left\|\frac{\overline{\mathbf{X}}_1^{*T}\overrightarrow{\mathbf{X}}}{\sqrt{n}}\right\|_2^2/\overrightarrow{p} \geq a\right) \leq \frac{\sum_{j=1}^{\overrightarrow{p}} E\left[\left(\frac{\overline{\mathbf{X}}_1^{*T}\overrightarrow{\mathbf{X}}_j}{\sqrt{n}}\right)^2\right]}{a\,\overrightarrow{p}} \leq C\frac{\sum_{j=2}^{p} E\left[\left(\frac{\overline{\mathbf{X}}_1^{*T}\mathbf{X}_j}{\sqrt{n}}\right)^2\right]}{a}, \tag{22}$$

where $C = \max_{j=2,\ldots,p}\left\{E\left[\bar{X}_{i,1}^{*2}\right]E\left[X_{i,j}^2\right]/E\left[\bar{X}_{i,1}^{*2}X_{i,j}^2\right]\right\} \vee 1$, and the moment in the right-hand side term converges by Assumption M and the von Bahr Theorem (Von Bahr, 1965, see also Theorem 6.3, DasGupta, 2008). The analogous argument holds for the norm $\left\|\overrightarrow{\mathbf{X}}^T\boldsymbol{\varepsilon}/\sqrt{n}\right\|_2$.

Taken together, we obtain (19). By Polya's Theorem, the convergence is uniform.

Hence, by Slutsky's theorem, it suffices to show that $\hat{\sigma}^2$ converges in probability to $\sigma^2 = E\left(\bar{X}_{1i}^{*2}\varepsilon_i^2\right)$. Consider the decomposition

$$\begin{aligned}\hat{\sigma}^2 - \sigma^2 &= \frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^2\hat{\varepsilon}_i^2 - E\left(\bar{X}_{i1}^{*2}\varepsilon_i^2\right)\\ &= \frac{1}{n}\sum_{i=1}^{n}\left(\bar{X}_{i1}^2 - \bar{X}_{i1}^{*2}\right)\left(\hat{\varepsilon}_i^2 - \varepsilon_i^2\right) + \frac{1}{n}\sum_{i=1}^{n}\left(\bar{X}_{i1}^2 - \bar{X}_{i1}^{*2}\right)\varepsilon_i^2\end{aligned}$$

[7] Remark that the scalar population analog is well defined because the vector population analog $\overline{\mathbf{X}}_1^* = \mathbf{X}_1 - E^*\left[\mathbf{X}_1\middle|\overrightarrow{\mathbf{X}}\right] = \mathbf{X}_1 - E^*\left[\mathbf{X}_1\middle|\mathbf{X}_2,\ldots,\mathbf{X}_p\right]$ has iid entries $\bar{X}_{i1}^* = X_{i1} - E^*\left[X_{i1}\middle|X_{i2},\ldots,X_{ip}\right]$.

$$+\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^{*2}\left(\hat{\varepsilon}_i^2-\varepsilon_i^2\right)+\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^{*2}\varepsilon_i^2-E\left(\bar{X}_{i1}^{*2}\varepsilon_i^2\right).$$

The first term satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\left(\bar{X}_{i1}^2-\bar{X}_{i1}^{*2}\right)\left(\hat{\varepsilon}_i^2-\varepsilon_i^2\right)\leq\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\bar{X}_{i1}^2-\bar{X}_{i1}^{*2}\right)^2}\cdot\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\hat{\varepsilon}_i^2-\varepsilon_i^2\right)^2}.$$

Both right-hand side terms can be treated analogously. Notice that

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^{n}\left(\bar{X}_{i1}^2-\bar{X}_{i1}^{*2}\right)^2&=\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^4-2\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^{*2}\bar{X}_{i1}^2+\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^{*4}\\&=E\left[\bar{X}_{i1}^{*4}\right]-2E\left[\bar{X}_{i1}^{*4}\right]+E\left[\bar{X}_{i1}^{*4}\right]+o_P(1),\end{aligned}$$

by M.2 and the law of large numbers. In particular, $\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^4\xrightarrow{P}E\left[\bar{X}_{i1}^{*4}\right]$ by M.2 and since $\|\hat{\gamma}-\gamma\|_2=O_P\left(\sqrt{\frac{\vec{p}^3}{n}}\right)$, which obtains by manipulations analogous to those in (21).

By a similar argument, the second term satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\left(\bar{X}_{i1}^2-\bar{X}_{i1}^{*2}\right)\varepsilon_i^2\leq\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\bar{X}_{i1}^2-\bar{X}_{i1}^{*2}\right)^2}\sqrt{\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^4}\xrightarrow{P}0.$$

By a symmetric argument, $\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^{*2}\left(\hat{\varepsilon}_i^2-\varepsilon_i^2\right)\xrightarrow{P}0$. Finally, $\frac{1}{n}\sum_{i=1}^{n}\bar{X}_{i1}^{*2}\varepsilon_i^2-E\left(\bar{X}_{i1}^{*2}\varepsilon_i^2\right)\xrightarrow{P}0$ by M.2 and the law of large numbers. *Q.E.D.*

We now turn to the large sample theory of the randomized test.

THEOREM A.3: *Let g and g' be independent and uniformly distributed over $\mathbf{G}_n$, i.e., $g,g'\sim U\left[\mathbf{G}_n\right]$, and be independent of the data $\mathbf{W}_n$ and let $\hat{R}_{T_g,n}$ be the permutation distribution of $T_n\left(g\mathbf{W}_n\right)$. Suppose, under $\mathcal{P}_n$,*

$$\left(T_n\left(g\mathbf{W}_n\right),T_n\left(g'\mathbf{W}_n\right)\right)\xrightarrow{d}\left(T,T'\right),\tag{23}$$

where T and T' are independent, each with common cumulative distribution function $R_T(\cdot)$. Then, for all continuity points S of $R_T(\cdot)$,

$$\hat{R}_{T_g,n}(S)\xrightarrow{P}R_T(S).\tag{24}$$

Conversely, if (24) holds for some limiting cumulative distribution function $R_T(\cdot)$ whenever S is a continuity point, then (23) holds.

The sufficiency extension of Hoeffding's Theorem is due to Chung and Romano (2013).

THEOREM A.4: *Suppose $(Y_i,X_i)\sim\mathcal{P}$ independently, and that Assumptions M, S and D hold. Suppose that $n_{\text{blocks}}^8/n\to0$. Further suppose that $\varepsilon_i^2=o_P(n)$, uniformly, and that*

$\frac{n_{\text{blocks}}}{b^{\zeta}} \to \infty$, *for some* $0 < \zeta < 1$. *Then, under the null hypothesis (3), the distribution* $\hat{R}_{t_g,n}$ *satisfies*

$$\hat{R}_{t_g,n}(s) \xrightarrow{P} \Phi(s),$$

for all points of continuity of s, *where* $\Phi(\cdot)$ *is the standard normal cumulative distribution.*

PROOF OF THEOREM A.4: Let

$$T_g = T(\mathbf{X}_1, \mathbf{X}_2, g(\mathbf{Y} - \mathbf{X}_1\beta_1)) = \frac{1}{\sqrt{n}}\overline{\mathbf{X}}_1^T g(\mathbf{Y} - \mathbf{X}_1\beta_1),$$

which under the null takes on the value

$$\frac{1}{\sqrt{n}}\overline{\mathbf{X}}_1^T (g\boldsymbol{\varepsilon}).$$

First, we want to verify Hoeffding's condition for T_g. Let $g, g' \sim U[\mathbf{G}_n]$, drawn independently, and consider the random pair $(T_g, T_{g'})$. We are interested in the limiting distribution of this bivariate random variable. We use the Cramér-Wold device. For any $(a_1, a_2) \in \mathbb{R}^2$, under the null hypothesis,

$$\begin{aligned} a_1 T_g + a_2 T_{g'} &= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \bar{X}_{i1} (a_1 \varepsilon_{g(i)} + a_2 \varepsilon_{g'(i)}) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \bar{X}^*_{i1} (a_1 \varepsilon_{g(i)} + a_2 \varepsilon_{g'(i)}) + o_P(1), \end{aligned}$$

by the same argument as in the proof of Theorem A.2.

This rewrites as

$$\frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_1} \bar{X}^*_{i1} (a_1 \varepsilon_{g(i)} + a_2 \varepsilon_{g'(i)}) + \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_1^c} \bar{X}^*_{i1} (a_1 \varepsilon_{g(i)} + a_2 \varepsilon_{g'(i)}) + o_P(1), \tag{25}$$

where $\mathcal{I}_1 = \{i : i \neq g(i) \wedge i \neq g'(i) \wedge g(i) \neq g'(i)\}$ and $\mathcal{I}_1^c = \{1, \ldots, n\} \setminus \mathcal{I}_1$.

Remark that $\frac{|\mathcal{I}_1|}{n} \xrightarrow{P} 1$. Indeed, using Markov's inequality,

$$P(|\mathcal{I}_1^c| \geq n_{\text{eq}}) \leq \frac{E[|\mathcal{I}_1^c|]}{n_{\text{eq}}} \leq \frac{3b}{n_{\text{eq}}}. \tag{26}$$

We can pick n_{eq} such that $\frac{n_{\text{eq}}}{n} \to 0$ and $\frac{b}{n_{\text{eq}}} \to 0$, as $n \to \infty$. We pick $n_{\text{eq}} = b^{1+\zeta}$.

We establish the CLT for the first summand of (25) by verifying the main condition of Lemma 11.3.3 of Lehmann et al. (2005), i.e.,

$$\frac{\max_{i \in \mathcal{I}_1} \frac{1}{|\mathcal{I}_1|} (a_1 \varepsilon_{g(i)} + a_2 \varepsilon_{g'(i)})^2}{\frac{1}{|\mathcal{I}_1|} \sum_{i \in \mathcal{I}_1} (a_1 \varepsilon_{g(i)} + a_2 \varepsilon_{g'(i)})^2} \xrightarrow{P} 0 \text{ as } |\mathcal{I}_1| \to \infty.$$

The probability limit of the denominator obtains as follows,

$$\frac{1}{|\mathcal{I}_1|}\sum_{i\in\mathcal{I}_1}\left(a_1\varepsilon_{g(i)}+a_2\varepsilon_{g'(i)}\right)^2$$

$$=\frac{a_1^2}{|\mathcal{I}_1|}\sum_{i\in\mathcal{I}_1}\varepsilon_{g(i)}^2+\frac{a_2^2}{|\mathcal{I}_1|}\sum_{i\in\mathcal{I}_1}\varepsilon_{g'(i)}^2+\frac{a_1a_2}{|\mathcal{I}_1|}\sum_{i\in\mathcal{I}_1}\varepsilon_{g(i)}\varepsilon_{g'(i)}$$

$$\xrightarrow{P} a_1^2E\left[\varepsilon_{g(i)}^2\right]+a_2^2E\left[\varepsilon_{g'(i)}^2\right]+0,$$

where we invoked Assumption M and the independence of $\varepsilon_{g(i)}$ and $\varepsilon_{g'(i)}$. Because $\varepsilon_i^2/n = o_P(1)$ uniformly, its numerator is $o_p(1)$. Therefore, by Slutsky's Theorem, the quotient must be $o_P(1)$, hence verifying the condition. Consequently,

$$\frac{1}{\sqrt{|\mathcal{I}_1|}}\sum_{i\in\mathcal{I}_1}\bar{X}_{i1}^*\left(a_1\varepsilon_{g(i)}+a_2\varepsilon_{g'(i)}\right)\xrightarrow{d}N\left(0,E\left[\bar{X}_{i1}^{*2}\right]\left(a_1^2E\left[\varepsilon_i^2\right]+a_2^2E\left[\varepsilon_i^2\right]\right)\right),$$

where we invoked the independence of $\bar{X}_{i1}^*$ from $\varepsilon_{g(i)}$ and $\varepsilon_{g'(i)}$ to produce the form of the asymptotic variance.

Consequently, the first term of (25) converges in distribution to $N\left(0,E\left[\bar{X}_{i1}^{*2}\right]\left(a_1^2E\left[\varepsilon_i^2\right]+a_2^2E\left[\varepsilon_i^2\right]\right)\right)$. The second term of (25) vanishes because $|\mathcal{I}_1^c|^{-1/2}\sum_{i\in\mathcal{I}_1^c}\bar{X}_{i1}^*\left(a_1\varepsilon_{g(i)}+a_2\varepsilon_{g'(i)}\right)=O_P(1)$ and $|\mathcal{I}_1^c|/n=o_p(1)$.

By Slutsky's Theorem, it obtains that

$$a_1T_g+a_2T_{g'}\xrightarrow{d}N\left(0,E\left[\bar{X}_{i1}^{*2}\right]\left(a_1^2E\left[\varepsilon_i^2\right]+a_2^2E\left[\varepsilon_i^2\right]\right)\right),$$

thus verifying Hoeffding's condition. In particular, we find that the asymptotic covariance between T_g and $T_{g'}$ is zero. By Hoeffding's Theorem, we then have that

$$\hat{R}_{T_g,n}(S)\xrightarrow{P}R_T(S), \tag{27}$$

for all points of continuity S, where $R_T(S)$ is the distribution function of $N\left(0,E\left[\bar{X}_{i1}^2\right]E\left[\varepsilon_i^2\right]\right)$.

Second, we verify that the studentizing term $\hat{\sigma}_g$ is consistent. Observe that

$$\begin{aligned}\hat{\sigma}_g^2-E\left[\bar{X}_{i1}^{*2}\right]E\left[\varepsilon_i^2\right]&=\frac{1}{n}\sum_{i=1}^n\bar{X}_{g(i),1}^2\hat{\varepsilon}_i^2-E\left[\bar{X}_{i1}^{*2}\right]E\left[\varepsilon_i^2\right]\\&=\frac{1}{n}\sum_{i:i\neq g(i)}\bar{X}_{g(i),1}^2\hat{\varepsilon}_i^2-E\left[\bar{X}_{i1}^{*2}\right]E\left[\varepsilon_i^2\right]+\frac{1}{n}\sum_{i:i=g(i)}\bar{X}_{g(i),1}^2\hat{\varepsilon}_i^2\\&=\frac{1}{n}\sum_{i:i\neq g(i)}\bar{X}_{g(i),1}^2\hat{\varepsilon}_i^2-E\left[\bar{X}_{i1}^{*2}\right]E\left[\varepsilon_i^2\right]+o_P(1)\\&=O(1)\left(\frac{1}{n-m}\sum_{i:i\neq g(i)}\bar{X}_{g(i),1}^2\varepsilon_i^2-E\left[\bar{X}_{i1}^{*2}\right]E\left[\varepsilon_i^2\right]\right)+o_P(1),\end{aligned}$$

where the number of matches is $m = \#\{i : i = g(i)\}$ and $m/n = o_P(1)$, and which goes to zero in probability by an argument analogous to that used in (26).

We must then have that

$$\hat{\sigma}_g^2 \xrightarrow{P} E\left[\bar{X}_{i1}^{*2}\right] E\left[\varepsilon_i^2\right]. \tag{28}$$

Third, we find by Slutsky's Theorem for randomization distributions, Theorem 5.2 of Chung and Romano (2013), that (27) and (28) imply

$$\hat{R}_{t_g,n}(s) \xrightarrow{P} R_t(s),$$

for all points of continuity of s, where $R_t(s)$ is the distribution function of $N(0,1)$. *Q.E.D.*

REMARK 1: We analyzed the case for a null hypothesis on a scalar coefficient. The large-sample theory for the multivariate case obtains analogously.

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